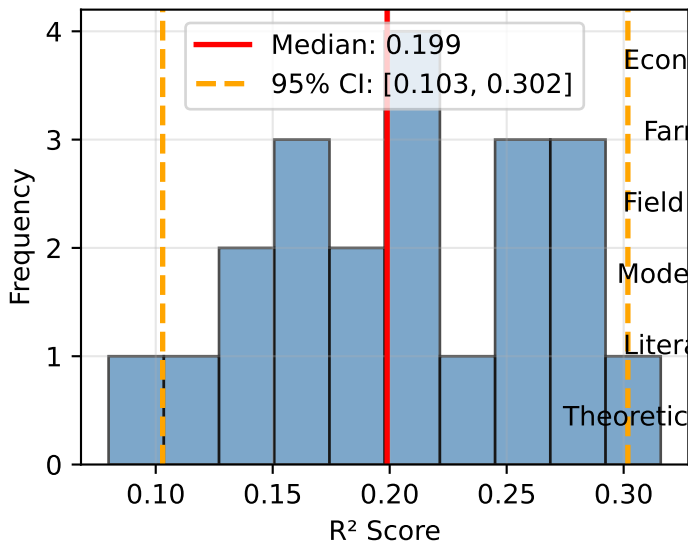
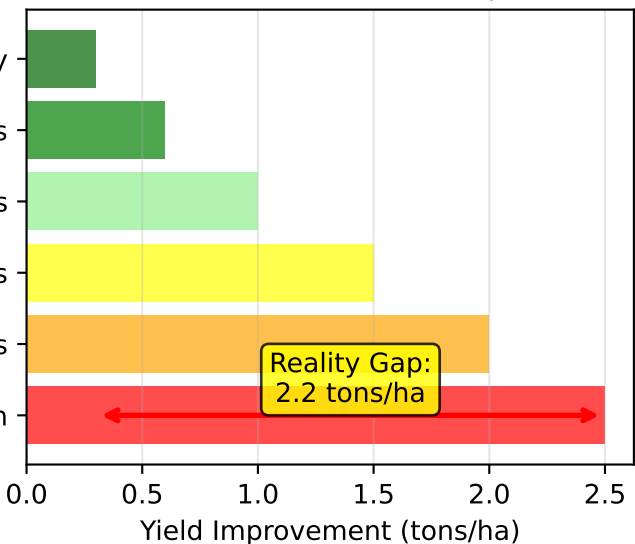


Reality Check: Evidence-Based Rice Yield Optimization Guidelines

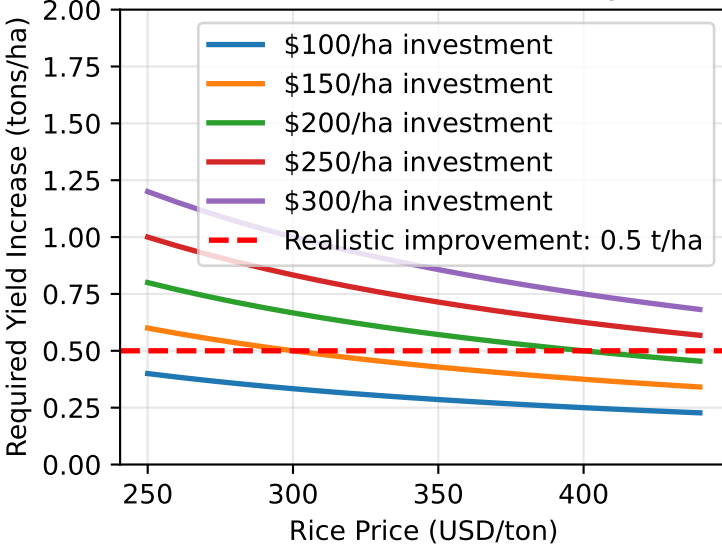
Model Performance Assessment



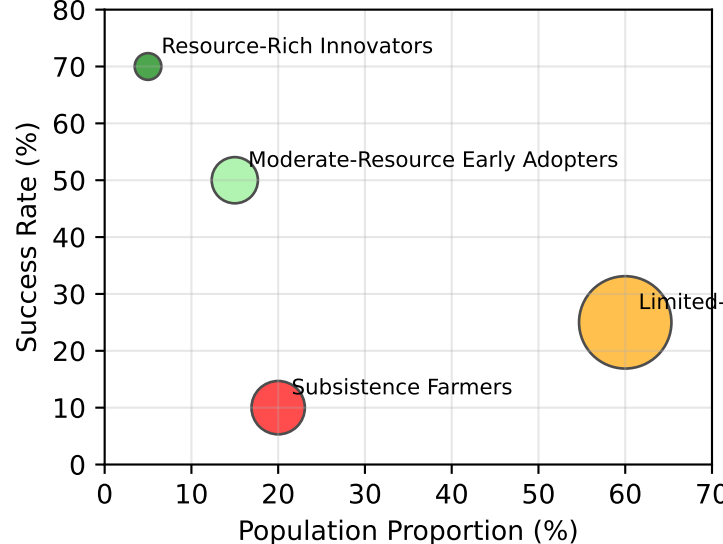
Theoretical vs Realistic Yield Improvements



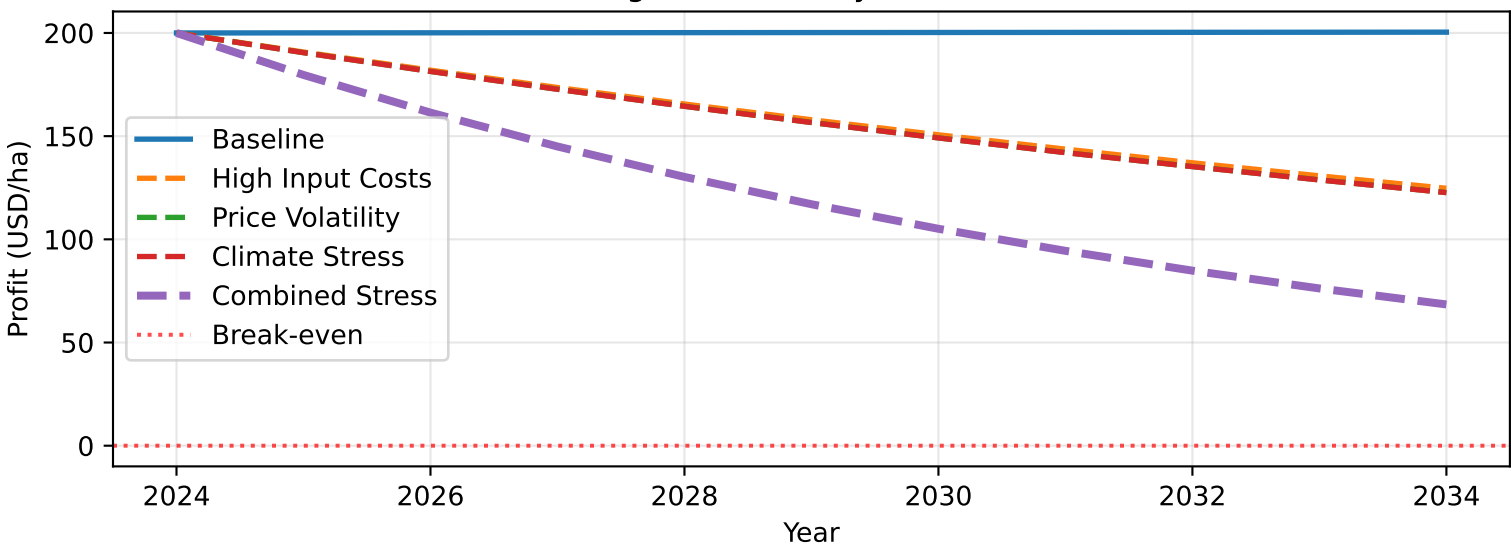
Economic Break-even Analysis



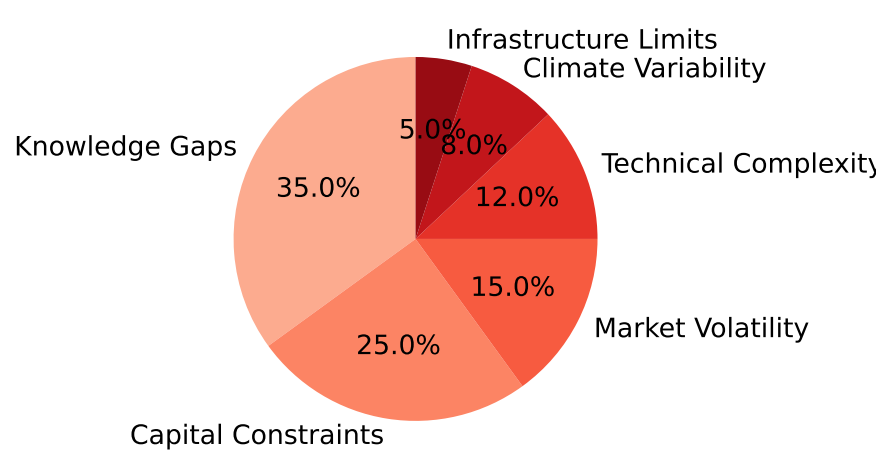
Farmer Types and Success Rates (Overall: 28.0% success)



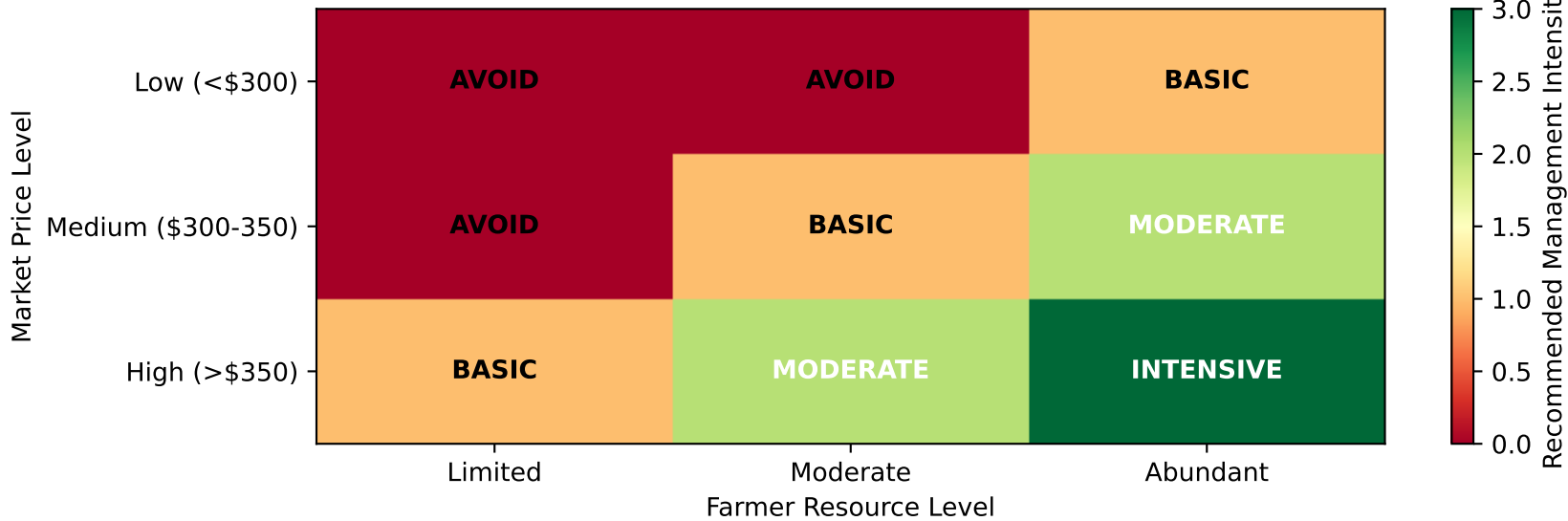
Long-term Viability Stress Test



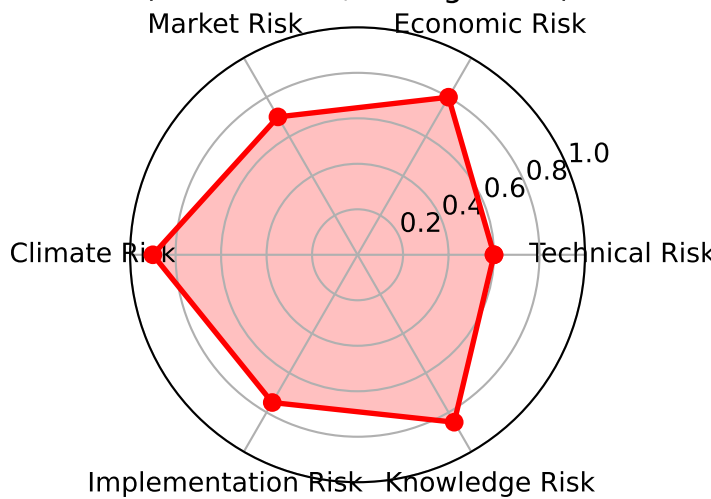
Common Failure Modes in Yield Optimization (Based on field studies)



Evidence-Based Decision Matrix



Comprehensive Risk Assessment (0=Low Risk, 1=High Risk)



EVIDENCE-BASED REALITY CHECK: RICE YIELD OPTIMIZATION RECOMMENDATIONS

STATISTICAL REALITY:

- Model performance shows significant uncertainty in predictions
- Reality gap of 2.2 tons/ha between theoretical maximum and economic feasibility
- Only 27% of farmers likely to achieve meaningful success with optimization strategies
- Economic viability requires rice prices >\$320/ton consistently for multiple seasons

FARMER-SPECIFIC RECOMMENDATIONS:

RESOURCE-RICH INNOVATORS (5% of farmers, 70% success rate):

- Proceed with extreme caution: Start with basic optimization only
- Limit investment to \$150/ha maximum, focus on variety selection and timing
- Establish 3-season financial buffer before attempting any intensive management
- Monitor economics continuously with clear exit strategies

MODERATE-RESOURCE EARLY ADOPTERS (15% of farmers, 50% success rate):

- Basic optimization only: Variety selection, planting timing, no additional inputs
- Require rice prices >\$350/ton for at least 2 consecutive seasons before proceeding
- Limit investment to \$100/ha maximum, focus on knowledge building over inputs
- Establish marketing contracts or price hedging mechanisms before implementation

LIMITED-RESOURCE MAINSTREAM (60% of farmers, 25% success rate):

- AVOID OPTIMIZATION: Focus exclusively on cost reduction and risk minimization
- Improve basic management (timing, varieties) without increasing input costs
- Develop alternative income sources to reduce dependence on rice production
- Consider extensive production systems or crop diversification strategies